

Statistical and Machine Learning SS26

Exercise Sheet 5

1) Kernel

- a) Load and inspect the dataset “Melbourne House Prices”.
- b) Plot the histogram of the distance from the city center as frequency and as density.
- c) Estimate the density of the Distance variable using four different kernel functions: Gaussian, Epanechnikov, Triangular, and Biweight. Use a fixed bandwidth of $bw = 1$ for all kernels. Plot all four curves on the same histogram. Why is using kernel density instead of regular histograms useful? What do you observe about the effect of kernel choice?
- d) Refit and plot the KDE using only the Gaussian kernel with three different bandwidths: $bw = 0.5$, $bw = 1$, and $bw = 1.5$. What’s the effect of bandwidth on smoothness and accuracy?
- e) Use `bw.bcv()` to select the optimal bandwidth automatically and estimate the probability of finding a house for sale between 8 and 10km from the CBD

2) K-Nearest Neighbor

The Caravan data set includes 85 predictors that measure demographic characteristics for 5,822 individuals. The goal is to predict whether someone buys caravan insurance. The response variable is **Purchase**, which indicates whether or not a given individual purchases a caravan insurance policy. In this data set, only 6 % of people purchased caravan insurance.

- a) Load and inspect the data (needed packages: `caret` and `ISLR2`)
- b) Why is standardization necessary before applying KNN? Standardize the predictor matrix using `scale()`, excluding the response variable **Purchase**.
- c) Split the data into a test set (observations 1–1000) and training set. Fit a KNN model with $K = 1$ on the training data and predict on the test set. What is the overall error rate? Compare it to the naive baseline of always predicting “No” - What do you observe and why?
- d) Compute the confusion matrix for $K = 1$. What is the precision among predicted buyers? Why is precision more informative than overall accuracy here?
- e) Refit the KNN model with $K = 3$ and $K = 5$. For each value report the confusion matrix and precision among predicted buyers. How does increasing K affect the number of predicted buyers and the precision? Why is higher precision cost-effective in this business context?

3) Gaussian Process Regression

Fit a Gaussian Process (GP) regression model to simulated data using the `gausspr()` function from the `kernlab` package and interpret the posterior mean and uncertainty.

What's a GP? -> A distribution over functions. Instead of predicting a single value like linear regression, it predicts 1) A mean: the best guess at each point 2) An uncertainty: how confident it is at each point -> plot: line with confidence band around it

Run the following code to generate the data.

```
library(kernlab)
library(ggplot2)

set.seed(42)
f_true <- function(x) sin(2 * pi * x) + 0.5 * sin(4 * pi * x)
n <- 40
x_train <- sort(runif(n, 0, 1))
y_train <- f_true(x_train) + rnorm(n, sd = 0.3)
x_grid <- seq(0, 1, length.out = 300)
```

- Fit a GP regression model using `gausspr()` with an RBF kernel (`kernel = "rbfdot"`), `sigma = 5`, and `var = 0.09`. Predict on `x_grid` and plot the training data together with the posterior mean and the true function (dashed line).
- Refit the GP with `variance.model = TRUE`. Extract the posterior standard deviations using `predict(..., type = "sdeviation")` and add ± 1.96 credible bands to your plot. Where is the uncertainty band widest and why?
- Refit the GP with `sigma = 1` and `sigma = 50` (keeping `var = 0.09`). Plot both posterior means together with the training data and the true function. What effect does `sigma` have on the fit?
- Using `sigma = 5`, refit the GP with `var = 0.001` and `var = 2`. Plot the two posterior means. What happens when `var` is too small? Too large? Why does `var = 0.09` give the best fit for this dataset?